

CSCE 411-501 ✖ Spring 2026

Undecidability Practice

Friday, January 23, 2026

1. Prove that the following languages are undecidable:

- (a) HALTS-EMPTY $:= \{\langle P \rangle : P(\varepsilon) \text{ halts}\}$
- (b) ALMOST-EMPTY $:= \{\langle P \rangle : P \text{ is an algorithm and } |L(P)| \leq 411\}$
- (c) COMPLEMENT $:= \{\langle P_1, P_2 \rangle : P_1, P_2 \text{ are algorithms \& } L(P_1) = \Sigma^* \setminus L(P_2)\}$
- (d) EVEN-DECIDER $:= \{\langle P \rangle : P \text{ is a decider for } \{\langle n \rangle : n \text{ is an even integer}\}\}$

2. Let L be a language with $|L| = 1$. Prove that L is decidable.

3. Show that the following language is undecidable:

DPCP $:= \{\langle P, d \rangle : P \text{ is an instance of the Post correspondence problem, } d \text{ is a domino in that instance, and } P \text{ has a solution which does not use the domino } d.\}$

For example, Suppose P is the the following set of dominos,

$$\left\{ \begin{bmatrix} b \\ ca \end{bmatrix}, \begin{bmatrix} a \\ ab \end{bmatrix}, \begin{bmatrix} bb \\ a \end{bmatrix}, \begin{bmatrix} ca \\ a \end{bmatrix}, \begin{bmatrix} abc \\ c \end{bmatrix} \right\},$$

Then $\langle P, \begin{bmatrix} bb \\ a \end{bmatrix} \rangle \in \text{DPCP}$, because P has the following solution,

$$\begin{bmatrix} a \\ ab \end{bmatrix} \begin{bmatrix} b \\ ca \end{bmatrix} \begin{bmatrix} ca \\ a \end{bmatrix} \begin{bmatrix} a \\ ab \end{bmatrix} \begin{bmatrix} abc \\ c \end{bmatrix},$$

which does not use the domino

$$\begin{bmatrix} bb \\ a \end{bmatrix}$$

4. Show that the function $f : \{P \mid P \text{ is an algorithm}\} \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ defined by $f(P, n) := \max_{x \in \Sigma^n \text{ s.t. } P(x) \text{ halts}} \text{the number of steps that } P(x) \text{ runs for before halting.}$ is uncomputable.